

PostgreSQL Query Plans

A practical reading guide for scans, joins, estimates and index decisions.

Read the plan as a tree

The most deeply indented operations generally run first. Each parent consumes rows from its child nodes. Look at row counts and loops before focusing on one large cost number.

EXPLAIN versus EXPLAIN ANALYZE

EXPLAIN shows the optimizer's estimates. EXPLAIN ANALYZE executes the query and adds actual timing and row counts. Use the latter carefully on statements that change data.

Sequential and index scans

A sequential scan is not automatically bad. It can be cheaper when a query needs much of a small table. Index scans become valuable when predicates are selective enough to avoid substantial work.

Composite indexes

Column order matters. An index beginning with `user_id` naturally supports predicates that constrain `user_id` before ranging or ordering on `created_at`.

Estimation mistakes

Large differences between estimated and actual rows can cause poor join or scan choices. Check statistics, data distribution and correlated columns.

Optimization checklist

Measure the real query, inspect buffers and rows, verify predicate selectivity, review existing indexes, then change one thing and measure again.

Plan signal	Question to ask	Possible response
Actual rows far above estimate	Are statistics representative?	Analyze table; inspect skew
Many rows removed by filter	Can predicate narrow earlier?	Review index and query
High loop count	Is nested work repeated?	Review join strategy
Sequential scan	Is a large share needed?	May already be optimal